

B5 General Relativity & Cosmology — Model Solutions, 2019

Question 1 — Equivalence principle and geodesic equation

(a) Equivalence principle and proper time

Problem. State the equivalence principle. In a free-falling lab with coordinates ξ^α , a particle obeys $d^2\xi^\alpha/d\tau^2 = 0$. Define the proper time τ .

Equivalence principle (strong form): at every spacetime event one can choose a local inertial (free-falling) frame in which the laws of physics take their special-relativistic form, and gravitational effects are locally indistinguishable from acceleration.

In the free-falling frame the metric is $\eta_{\alpha\beta} = \text{diag}(-1, +1, +1, +1)$. The invariant proper time is

$$c^2 d\tau^2 = -\eta_{\alpha\beta} d\xi^\alpha d\xi^\beta.$$

Equivalently:

$$d\tau^2 = -\frac{1}{c^2} \eta_{\alpha\beta} d\xi^\alpha d\xi^\beta.$$

(b) Transformation to general coordinates

Problem. Starting from $d^2\xi^\alpha/d\tau^2 = 0$, derive the geodesic equation in coordinates x^α and identify the symmetry of Γ .

Chain rule on $\xi^\alpha = \xi^\alpha(x^\mu)$:

$$\frac{d\xi^\alpha}{d\tau} = \frac{\partial \xi^\alpha}{\partial x^\mu} \frac{dx^\mu}{d\tau}.$$

Differentiate again with the product rule:

$$\frac{d^2\xi^\alpha}{d\tau^2} = \frac{\partial \xi^\alpha}{\partial x^\mu} \frac{d^2x^\mu}{d\tau^2} + \frac{\partial^2 \xi^\alpha}{\partial x^\mu \partial x^\nu} \frac{dx^\mu}{d\tau} \frac{dx^\nu}{d\tau} = 0.$$

Multiply by $\partial x^\lambda / \partial \xi^\alpha$:

$$\frac{\partial x^\lambda}{\partial \xi^\alpha} \frac{\partial \xi^\alpha}{\partial x^\mu} \frac{d^2x^\mu}{d\tau^2} + \frac{\partial x^\lambda}{\partial \xi^\alpha} \frac{\partial^2 \xi^\alpha}{\partial x^\mu \partial x^\nu} \frac{dx^\mu}{d\tau} \frac{dx^\nu}{d\tau} = 0.$$

Apply $(\partial x^\lambda / \partial \xi^\alpha)(\partial \xi^\alpha / \partial x^\mu) = \delta_\mu^\lambda$:

$$\frac{d^2x^\lambda}{d\tau^2} + \frac{\partial x^\lambda}{\partial \xi^\alpha} \frac{\partial^2 \xi^\alpha}{\partial x^\mu \partial x^\nu} \frac{dx^\mu}{d\tau} \frac{dx^\nu}{d\tau} = 0.$$

Identify the connection:

$$\Gamma_{\mu\nu}^\lambda = \frac{\partial x^\lambda}{\partial \xi^\alpha} \frac{\partial^2 \xi^\alpha}{\partial x^\mu \partial x^\nu}.$$

Geodesic equation:

$$\frac{d^2x^\lambda}{d\tau^2} + \Gamma_{\mu\nu}^\lambda \frac{dx^\mu}{d\tau} \frac{dx^\nu}{d\tau} = 0.$$

Since partial derivatives commute, $\Gamma_{\mu\nu}^\lambda = \Gamma_{\nu\mu}^\lambda$ (symmetric in lower indices).

(c) Metric compatibility relation

Problem. With $g_{\mu\nu} = \eta_{\alpha\beta}(\partial\xi^\alpha/\partial x^\mu)(\partial\xi^\beta/\partial x^\nu)$, show $\partial_\lambda g_{\mu\nu} = g_{\rho\nu}\Gamma_{\lambda\mu}^\rho + g_{\mu\rho}\Gamma_{\lambda\nu}^\rho$.

Invariant line element:

$$ds^2 = \eta_{\alpha\beta} d\xi^\alpha d\xi^\beta = g_{\mu\nu} dx^\mu dx^\nu, \quad g_{\mu\nu} = \eta_{\alpha\beta} \frac{\partial\xi^\alpha}{\partial x^\mu} \frac{\partial\xi^\beta}{\partial x^\nu}.$$

Differentiate $g_{\mu\nu}$:

$$\frac{\partial g_{\mu\nu}}{\partial x^\lambda} = \eta_{\alpha\beta} \frac{\partial^2 \xi^\alpha}{\partial x^\lambda \partial x^\mu} \frac{\partial \xi^\beta}{\partial x^\nu} + \eta_{\alpha\beta} \frac{\partial \xi^\alpha}{\partial x^\mu} \frac{\partial^2 \xi^\beta}{\partial x^\lambda \partial x^\nu}.$$

Insert $\partial^2 \xi^\alpha / \partial x^\lambda \partial x^\mu = \Gamma_{\lambda\mu}^\rho \partial \xi^\alpha / \partial x^\rho$ from the connection definition:

$$\frac{\partial g_{\mu\nu}}{\partial x^\lambda} = \eta_{\alpha\beta} \Gamma_{\lambda\mu}^\rho \frac{\partial \xi^\alpha}{\partial x^\rho} \frac{\partial \xi^\beta}{\partial x^\nu} + \eta_{\alpha\beta} \Gamma_{\lambda\nu}^\rho \frac{\partial \xi^\alpha}{\partial x^\mu} \frac{\partial \xi^\beta}{\partial x^\rho}.$$

Recognise $g_{\rho\nu}$ and $g_{\mu\rho}$:

$$\frac{\partial g_{\mu\nu}}{\partial x^\lambda} = g_{\rho\nu} \Gamma_{\lambda\mu}^\rho + g_{\mu\rho} \Gamma_{\lambda\nu}^\rho.$$

(d) Christoffel formula

Problem. Use the symmetry of Γ to express $\Gamma_{\mu\lambda}^\sigma$ in terms of $g_{\mu\nu}$.

Cyclic permutation of (λ, μ, ν) in the previous identity:

$$\partial_\mu g_{\nu\lambda} = g_{\rho\lambda} \Gamma_{\mu\nu}^\rho + g_{\nu\rho} \Gamma_{\mu\lambda}^\rho,$$

$$\partial_\nu g_{\lambda\mu} = g_{\rho\mu} \Gamma_{\nu\lambda}^\rho + g_{\lambda\rho} \Gamma_{\nu\mu}^\rho.$$

Compute $\partial_\mu g_{\nu\lambda} + \partial_\nu g_{\lambda\mu} - \partial_\lambda g_{\mu\nu}$ and use $\Gamma_{\mu\nu}^\rho = \Gamma_{\nu\mu}^\rho$:

$$\partial_\mu g_{\nu\lambda} + \partial_\nu g_{\lambda\mu} - \partial_\lambda g_{\mu\nu} = 2 g_{\rho\lambda} \Gamma_{\mu\nu}^\rho.$$

Multiply by $\frac{1}{2} g^{\sigma\lambda}$ and relabel:

$$\Gamma_{\mu\lambda}^\sigma = \frac{1}{2} g^{\nu\sigma} \left(\frac{\partial g_{\mu\nu}}{\partial x^\lambda} + \frac{\partial g_{\lambda\nu}}{\partial x^\mu} - \frac{\partial g_{\lambda\mu}}{\partial x^\nu} \right).$$

(e) Covariant derivative

Problem. Define $U^\lambda = dx^\lambda/d\tau$. Use the geodesic equation to define the covariant derivative; comment on its rank and significance.

Geodesic equation rewritten using U^λ :

$$\frac{dU^\lambda}{d\tau} + \Gamma_{\mu\nu}^\lambda U^\mu U^\nu = 0.$$

Use $dU^\lambda/d\tau = U^\nu \partial_\nu U^\lambda$:

$$U^\nu \left(\partial_\nu U^\lambda + \Gamma_{\mu\nu}^\lambda U^\mu \right) = 0.$$

This identifies the covariant derivative:

$$\nabla_\nu U^\lambda \equiv \partial_\nu U^\lambda + \Gamma_{\mu\nu}^\lambda U^\mu.$$

Significance. $\nabla_\nu U^\lambda$ corrects the partial derivative for the rotation/stretching of the coordinate basis induced by curvature, so that geodesic motion ($\nabla_U U = 0$) is the curved-space analogue of Newton's first law: a free particle parallel-transport its own four-velocity.

Tensor character. Under $x^\mu \rightarrow x'^\mu$:

$$\begin{aligned}\partial_\nu U^\lambda &\rightarrow \frac{\partial x^\beta}{\partial x'^\nu} \left(\frac{\partial x'^\lambda}{\partial x^\alpha} \partial_\beta U^\alpha + U^\alpha \frac{\partial^2 x'^\lambda}{\partial x^\alpha \partial x^\beta} \right), \\ \Gamma_{\mu\nu}^\lambda U^\mu &\rightarrow \frac{\partial x'^\lambda}{\partial x^\alpha} \frac{\partial x^\beta}{\partial x'^\nu} \Gamma_{\gamma\beta}^\alpha U^\gamma - \frac{\partial x^\beta}{\partial x'^\nu} U^\alpha \frac{\partial^2 x'^\lambda}{\partial x^\alpha \partial x^\beta}.\end{aligned}$$

The non-tensorial second-derivative pieces cancel:

$$\nabla_\nu U^\lambda \rightarrow \frac{\partial x'^\lambda}{\partial x^\alpha} \frac{\partial x^\beta}{\partial x'^\nu} \nabla_\beta U^\alpha.$$

Hence $\nabla_\nu U^\lambda$ transforms as a (1,1) mixed rank-2 tensor: required so that $\nabla_U U = 0$ has covariant meaning.

Question 2 — Schwarzschild geodesics and orbits

(a) Geodesic equations from the Lagrangian

Problem. Write the geodesic equations for the Schwarzschild line element using the Lagrange procedure; explain the θ equation; relate p^α to the geodesics.

Effective Lagrangian (one-half line element):

$$\mathcal{L} = \frac{1}{2} g_{\mu\nu} \dot{x}^\mu \dot{x}^\nu = \frac{1}{2} \left[- \left(1 - \frac{2GM}{c^2 r} \right) c^2 \dot{t}^2 + \left(1 - \frac{2GM}{c^2 r} \right)^{-1} \dot{r}^2 + r^2 \dot{\theta}^2 + r^2 \sin^2 \theta \dot{\phi}^2 \right].$$

Euler–Lagrange equations $\frac{d}{d\tau} \partial_{\dot{x}^\mu} \mathcal{L} - \partial_{x^\mu} \mathcal{L} = 0$:

$$\begin{aligned}\frac{d}{d\tau} \left[\left(1 - \frac{2GM}{c^2 r} \right) c^2 \dot{t} \right] &= 0, \\ \frac{d}{d\tau} \left[\left(1 - \frac{2GM}{c^2 r} \right)^{-1} \dot{r} \right] - \frac{1}{2} \left[- \frac{2GM}{c^2 r^2} c^2 \dot{t}^2 - \frac{2GM}{c^2 r^2} \left(1 - \frac{2GM}{c^2 r} \right)^{-2} \dot{r}^2 + 2r \dot{\theta}^2 + 2r \sin^2 \theta \dot{\phi}^2 \right] &= 0, \\ \frac{d}{d\tau} \left[r^2 \dot{\theta} \right] - r^2 \sin \theta \cos \theta \dot{\phi}^2 &= 0, \\ \frac{d}{d\tau} \left[r^2 \sin^2 \theta \dot{\phi} \right] &= 0.\end{aligned}$$

Spherical symmetry: take initial $\theta = \pi/2$, $\dot{\theta} = 0$:

$$\ddot{\theta} = 0, \quad \theta(\tau) = \pi/2 \text{ identically.}$$

The orbit lies in the equatorial plane.

Four-momentum. For a massive particle $p^\alpha = m dx^\alpha/d\tau$; the geodesic equation is the parallel-transport law $p^\nu \nabla_\nu p^\alpha = 0$, i.e. free particles follow geodesic curves with their four-momentum tangent to the geodesic.

(b) Affine parameter and constants of motion

Problem. When is τ an appropriate affine parameter? Identify the conserved quantities k and h from the t and ϕ equations.

Proper time τ is affine for *massive* particles where $d\tau^2 = -ds^2/c^2 > 0$. For *massless* particles $d\tau = 0$, so use any affine parameter λ such that $dp^\alpha/d\lambda + \Gamma_{\mu\nu}^\alpha p^\mu p^\nu/m = 0$ remains well-defined (e.g. λ chosen so $p^\alpha = dx^\alpha/d\lambda$).

Integrate the t -equation:

$$\left(1 - \frac{2GM}{c^2 r}\right) c^2 \dot{t} = \text{const} \equiv c^2 k.$$

Integrate the ϕ -equation in the equatorial plane:

$$r^2 \dot{\phi} = \text{const} \equiv h.$$

With $p_\alpha = g_{\alpha\beta} p^\beta = m g_{\alpha\beta} \dot{x}^\beta$:

$$p_0 = -m \left(1 - \frac{2GM}{c^2 r}\right) c^2 \dot{t} = -mc^2 k, \quad p_3 = mr^2 \sin^2 \theta \dot{\phi} = mh.$$

Physical meaning. $k = -p_0/(mc^2)$ is the conserved *energy per unit rest mass* measured at infinity (Killing vector ∂_t); $h = p_3/m$ is the conserved *specific angular momentum* (Killing vector ∂_ϕ).

(c) Energy equation

Problem. From $g_{\mu\nu} \dot{x}^\mu \dot{x}^\nu = c^2$ derive the radial energy equation.

Insert the equatorial line element with $\dot{\theta} = 0$:

$$-\left(1 - \frac{2GM}{c^2 r}\right) c^2 \dot{t}^2 + \left(1 - \frac{2GM}{c^2 r}\right)^{-1} \dot{r}^2 + r^2 \dot{\phi}^2 = c^2.$$

Wait — the convention here uses $g_{\mu\nu} \dot{x}^\mu \dot{x}^\nu = c^2$ (the question's sign), so the line element with the question's metric signature $(-, +, +, +)$ multiplied by -1 gives $+c^2$. Substitute $c\dot{t} = ck/(1 - 2GM/c^2 r)$ and $\dot{\phi} = h/r^2$:

$$-\frac{c^2 k^2}{1 - 2GM/(c^2 r)} + \frac{\dot{r}^2}{1 - 2GM/(c^2 r)} + \frac{h^2}{r^2} = -c^2.$$

Multiply by $(1 - 2GM/(c^2 r))$:

$$-c^2 k^2 + \dot{r}^2 + \frac{h^2}{r^2} \left(1 - \frac{2GM}{c^2 r}\right) = -c^2 \left(1 - \frac{2GM}{c^2 r}\right).$$

Rearrange for $\dot{r}^2$:

$$\boxed{\dot{r}^2 + \frac{h^2}{r^2} \left(1 - \frac{2GM}{c^2 r}\right) - \frac{2GM}{r} = c^2 (k^2 - 1).}$$

(d) Orbit equation and relativistic correction

Problem. Using () and $r^2 \dot{\phi} = h$ derive $d^2 u/d\phi^2 + u = GM/h^2 + 3GMu^2/c^2$ with $u = 1/r$.

Change variable $u = 1/r$. Chain rule:

$$\dot{r} = \frac{dr}{d\phi} \dot{\phi} = -\frac{1}{u^2} \frac{du}{d\phi} \cdot \frac{h}{r^2} = -h \frac{du}{d\phi}.$$

Square:

$$\dot{r}^2 = h^2 \left(\frac{du}{d\phi} \right)^2.$$

Substitute into () and use $1/r = u$:

$$h^2 \left(\frac{du}{d\phi} \right)^2 + h^2 u^2 \left(1 - \frac{2GMu}{c^2} \right) - 2GMu = c^2 (k^2 - 1).$$

Differentiate w.r.t. ϕ :

$$2h^2 \frac{du}{d\phi} \frac{d^2u}{d\phi^2} + 2h^2 u \frac{du}{d\phi} - 6 \frac{GMh^2}{c^2} u^2 \frac{du}{d\phi} - 2GM \frac{du}{d\phi} = 0.$$

Divide by $2h^2 du/d\phi$ (non-circular orbit):

$$\boxed{\frac{d^2u}{d\phi^2} + u = \frac{GM}{h^2} + \frac{3GM}{c^2} u^2.}$$

The Newtonian equation has only the constant source GM/h^2 . The relativistic correction is the term $3GMu^2/c^2$. Its ratio to the Newtonian term is $3h^2u^2/c^2 = 3v_\perp^2/c^2$ where $v_\perp = hu$, so it is significant when $v \sim c$, i.e. for tight orbits at $r \sim r_s = 2GM/c^2$ (e.g. Mercury's perihelion advance, S2 around Sgr A*).

(e) Effective potential in GR

Problem. Cast the energy equation in Newtonian form $\frac{1}{2}(\dot{r})^2 + V_{\text{eff}}(r) = E$.

In the orbiting observer's frame the affine parameter is τ , so the Newtonian dr/dt is replaced by $dr/d\tau = \dot{r}$. From ():

$$\frac{1}{2}\dot{r}^2 + \frac{1}{2} \left[\frac{h^2}{r^2} \left(1 - \frac{2GM}{c^2 r} \right) - \frac{2GM}{r} \right] = \frac{1}{2}c^2(k^2 - 1).$$

Identify the effective potential:

$$\boxed{V_{\text{eff}}(r) = -\frac{GM}{r} + \frac{h^2}{2r^2} - \frac{GMh^2}{c^2 r^3}, \quad E = \frac{1}{2}c^2(k^2 - 1).}$$

The first two terms reproduce the Newtonian gravity + centrifugal barrier; the $1/r^3$ term is purely relativistic.

(f) Circular orbits

Problem. Show circular-orbit extrema occur at $r = (h/2GM) \left(h \pm \sqrt{h^2 - 12G^2M^2/c^2} \right)$.

Circular orbits: $dV_{\text{eff}}/dr = 0$. Differentiate ():

$$\frac{dV_{\text{eff}}}{dr} = \frac{GM}{r^2} - \frac{h^2}{r^3} + \frac{3GMh^2}{c^2 r^4}.$$

Set to zero and multiply by r^4 :

$$GM r^2 - h^2 r + \frac{3GMh^2}{c^2} = 0.$$

Quadratic in r :

$$r = \frac{h^2 \pm \sqrt{h^4 - 12G^2M^2h^2/c^2}}{2GM}.$$

Factor h from the surd:

$$r = \frac{h}{2GM} \left(h \pm \sqrt{h^2 - \frac{12G^2M^2}{c^2}} \right).$$

The $+$ root is the stable minimum of V_{eff} (regular orbit), the $-$ root is the unstable maximum (light-ring / unstable circular orbit). Real solutions require $h^2 \geq 12G^2M^2/c^2$; equality gives the innermost stable circular orbit (ISCO) at $r = 6GM/c^2$.

Question 3 — Binary pulsar and gravitational radiation

(a) Why a GR expert should notice

Problem. *Pulsar in a binary, $T = 0.102\,251\,562\,48$ day. Why does this matter for GR?*

A binary system containing a pulsar provides a clean clock embedded in a strong-field, dynamical spacetime: the orbit decays via emission of gravitational waves, and the highly accurate pulse timing lets one measure $\dot{T}$ at the level needed to test the quadrupole formula of GR (Hulse–Taylor / double pulsar). It is one of very few systems testing the radiative sector of Einstein’s theory.

(b) Orbital radius

Problem. $T_0 = 0.102\,251\,562\,48$ day, $\dot{T} \approx -10^{-9}$ day/1000 day, $m_1 = m_2 = 1.4M_\odot$, *circular orbit. Find a .*

Convert period to SI:

$$T = 0.10225156248 \times 86400 \text{ s} = 8834.535 \text{ s}.$$

Apply Kepler’s third law (total mass $M_{\text{tot}} = 2.8 M_\odot$, $GM_\odot = 1.327 \times 10^{20} \text{ m}^3 \text{ s}^{-2}$):

$$a^3 = \frac{GM_{\text{tot}}T^2}{4\pi^2}.$$

Numerically:

$$GM_{\text{tot}} = 2.8 \times 1.327 \times 10^{20} \text{ m}^3 \text{ s}^{-2} = 3.716 \times 10^{20} \text{ m}^3 \text{ s}^{-2}.$$

$$T^2 = (8834.535 \text{ s})^2 = 7.804 \times 10^7 \text{ s}^2.$$

$$GM_{\text{tot}}T^2 = 2.900 \times 10^{28} \text{ m}^3.$$

$$a^3 = \frac{2.900 \times 10^{28} \text{ m}^3}{4\pi^2} = 7.346 \times 10^{26} \text{ m}^3.$$

$$a \approx 9.02 \times 10^8 \text{ m} \approx 1.3 R_\odot.$$

(c) Definitions in the quadrupole formula

Problem. *Define $\bar{h}^{ij}$ and t' in $\bar{h}^{ij} = (2G/c^6 r) d^2 I^{ij} / dt'^2$.*

$\bar{h}^{\mu\nu}$ is the *trace-reversed* metric perturbation in the harmonic gauge:

$$g_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu}, \quad \bar{h}_{\mu\nu} = h_{\mu\nu} - \frac{1}{2}\eta_{\mu\nu} h, \quad h \equiv \eta^{\mu\nu} h_{\mu\nu}.$$

The harmonic-gauge condition $\partial_\mu \bar{h}^{\mu\nu} = 0$ reduces the linearised Einstein equation to $\square \bar{h}^{\mu\nu} = -(16\pi G/c^4) T^{\mu\nu}$, whose retarded solution gives the quadrupole formula.

t' is the *retarded time* at the source:

$$t' = t - r/c,$$

the time of emission for a wave received at the field point $(t, \mathbf{r})$.

(d) Meaning of x^i, x^j

Problem. What do x^i, x^j in $I^{ij} = Mc^2 x^i x^j$ represent, given that pulsars are point-like?

For two point masses orbiting their common centre of mass, the source quadrupole is concentrated on the orbit. Pulsars are ~ 10 km compared to $a \sim 10^9$ m, so each is treated as a point mass. With $M = 2M_p$ and equal masses on a circular orbit of radius $a/2$, the relative position $\mathbf{r}_{12} = \mathbf{x}_1 - \mathbf{x}_2$ has magnitude a . The reduced single-particle quadrupole becomes $I^{ij} = \mu r_{12}^i r_{12}^j$ with $\mu = M/4 = M_p/2$. The form $I^{ij} = Mc^2 x^i x^j$ in the question identifies x^i as a *coordinate of the relative orbital position* (one component of $\mathbf{r}_{12}/2$ or, equivalently, of the orbital trajectory) such that the prefactor Mc^2 reproduces the correct magnitude when the two equal point-pulsars are taken into account. They are *not* internal pulsar coordinates: pulsars are point-like at this scale.

(e) GW frequency and waveform

Problem. Find f_{GW} to 2 s.f.; write $\bar{h}^{ij}(t')$.

Orbital angular frequency:

$$\omega_{\text{orb}} = \frac{2\pi}{T} = \frac{2\pi}{8834.535 \text{ s}} = 7.111 \times 10^{-4} \text{ rad/s.}$$

Orbital frequency:

$$f_{\text{orb}} = 1/T = 1.132 \times 10^{-4} \text{ Hz.}$$

Quadrupole radiation has $f_{\text{GW}} = 2f_{\text{orb}}$:

$$f_{\text{GW}} = 2.3 \times 10^{-4} \text{ Hz.}$$

Waveform. Place the orbit in the xy -plane; equal masses give relative coordinate

$$x^1(t') = (a/2) \cos(\omega_{\text{orb}} t'), \quad x^2(t') = (a/2) \sin(\omega_{\text{orb}} t'), \quad x^3 = 0.$$

Form the components $I^{ij} = Mc^2 x^i x^j$ (with $M = 2M_p$):

$$I^{11} = Mc^2 (a/2)^2 \cos^2(\omega t') = \frac{1}{2} Mc^2 (a/2)^2 [1 + \cos(2\omega t')],$$

$$I^{22} = \frac{1}{2} Mc^2 (a/2)^2 [1 - \cos(2\omega t')],$$

$$I^{12} = Mc^2 (a/2)^2 \sin(\omega t') \cos(\omega t') = \frac{1}{2} Mc^2 (a/2)^2 \sin(2\omega t').$$

Differentiate twice (constants drop):

$$\ddot{I}^{11} = -2\omega^2 Mc^2 (a/2)^2 \cos(2\omega t') = -\frac{1}{2} Mc^2 a^2 \omega^2 \cos(2\omega t'),$$

$$\ddot{I}^{22} = +\frac{1}{2} Mc^2 a^2 \omega^2 \cos(2\omega t'),$$

$$\ddot{I}^{12} = -\frac{1}{2} Mc^2 a^2 \omega^2 \sin(2\omega t').$$

Insert in $\bar{h}^{ij} = (2G/c^6 r) \ddot{I}^{ij}$:

$$\bar{h}^{ij}(t') = \frac{G M a^2 \omega^2}{c^4 r} \begin{pmatrix} -\cos 2\omega t' & -\sin 2\omega t' & 0 \\ -\sin 2\omega t' & +\cos 2\omega t' & 0 \\ 0 & 0 & 0 \end{pmatrix},$$

oscillating at angular frequency $2\omega_{\text{orb}}$, hence frequency 2.3×10^{-4} Hz.

(f) Why $\dot{T}$ tests GR

Problem. *Why does measuring $\dot{T}$ test general relativity?*

Energy radiated as gravitational waves, summed over the two polarisations and integrated over the sphere, is given by the Einstein quadrupole formula

$$P_{\text{GW}} = \frac{G}{5c^5} \langle \ddot{I}_{\text{TT}}^{ij} \ddot{I}_{\text{TT}ij} \rangle.$$

For a circular binary of masses m_1, m_2 and separation a this evaluates to

$$P_{\text{GW}} = \frac{32}{5} \frac{G^4}{c^5} \frac{(m_1 m_2)^2 (m_1 + m_2)}{a^5}.$$

This luminosity drains the orbital binding energy $E_{\text{orb}} = -Gm_1 m_2 / (2a)$, shrinking a and (by Kepler) shortening T . Differentiating Kepler's third law gives a sharp prediction

$$\dot{T} = -\frac{192\pi}{5c^5} \left(\frac{2\pi G}{T} \right)^{5/3} \frac{m_1 m_2}{(m_1 + m_2)^{1/3}},$$

i.e. $\dot{T}$ is a definite number once both pulsar masses are known. The slight orbital eccentricity allows independent measurement of post-Keplerian parameters (periastron advance, Shapiro delay, gravitational redshift) which over-determine m_1, m_2 . Comparing the predicted $\dot{T}$ with the measured one with no free parameters is therefore a direct test of GR's radiative quadrupole sector — the test famously passed by the Hulse–Taylor system and by the double pulsar to better than 0.1%.

Question 4 — Cosmological constant and the FRW universe

(a) Bianchi identities

Problem. *State the contracted Bianchi identities and verify that the modified Einstein equation respects them. Why is this physically significant?*

Contracted Bianchi identity:

$$\nabla^\mu (R_{\mu\nu} - \tfrac{1}{2} g_{\mu\nu} R) = 0.$$

Take the covariant divergence of the modified equation:

$$\nabla^\mu (R_{\mu\nu} - \tfrac{1}{2} g_{\mu\nu} R) + \Lambda \nabla^\mu g_{\mu\nu} = -\frac{8\pi G}{c^4} \nabla^\mu T_{\mu\nu}.$$

Use $\nabla^\mu g_{\mu\nu} = 0$ (metric compatibility) and the Bianchi identity:

$$0 = -\frac{8\pi G}{c^4} \nabla^\mu T_{\mu\nu} \Rightarrow \nabla^\mu T_{\mu\nu} = 0.$$

Significance. The modified equation enforces local conservation of energy–momentum automatically. Adding $\Lambda g_{\mu\nu}$ does not spoil this because $g_{\mu\nu}$ is covariantly constant; any other proposed addition would have to share that property.

(b) Four-velocity of the cosmic fluid

Problem. *Define U^μ in $T_{\mu\nu} = P g_{\mu\nu} + (\rho + P/c^2) U_\mu U_\nu$. Give U^λ, U_λ at $\lambda = t, r$ in the FRW comoving frame.*

$U^\mu = dx^\mu/d\tau$ is the four-velocity of a fluid element measured in its own proper time, normalised $g_{\mu\nu} U^\mu U^\nu = -c^2$.

In the comoving frame the fluid is at rest in spatial coordinates: $dr = d\theta = d\phi = 0$. From $-c^2 d\tau^2 = -c^2 dt^2$ for a comoving worldline, $dt = d\tau$:

$$U^t = c, \quad U^r = 0.$$

(Equivalently, $U^\mu = (c, 0, 0, 0)$ in (ct, r, θ, ϕ) coordinates with $g_{tt} = -1$.)

Lower the index using $g_{tt} = -1, g_{rr} = R^2/(1 - r^2/a^2)$:

$$U_t = g_{tt}U^t = -c, \quad U_r = g_{rr}U^r = 0.$$

$U^t = c, \quad U^r = 0; \quad U_t = -c, \quad U_r = 0.$

(c) Effective fluid with Λ

Problem. Show that with $\rho \rightarrow \rho', P \rightarrow P'$ the modified equation reduces to the original; evaluate ρ_V and interpret.

Move the Λ -term to the right of the modified equation:

$$R_{\mu\nu} - \frac{1}{2}g_{\mu\nu}R = -\frac{8\pi G}{c^4}T_{\mu\nu} - \Lambda g_{\mu\nu}.$$

Express the extra piece as a fluid contribution:

$$-\Lambda g_{\mu\nu} = -\frac{8\pi G}{c^4}T_{\mu\nu}^{(V)} \Rightarrow T_{\mu\nu}^{(V)} = \frac{\Lambda c^4}{8\pi G}g_{\mu\nu}.$$

Match to perfect-fluid form $T_{\mu\nu}^{(V)} = P_V g_{\mu\nu} + (\rho_V + P_V/c^2)U_\mu U_\nu$ with $P_V = -\rho_V c^2$:

$$T_{\mu\nu}^{(V)} = -\rho_V c^2 g_{\mu\nu}.$$

Equate:

$$-\rho_V c^2 = \frac{\Lambda c^4}{8\pi G} \Rightarrow \boxed{\rho_V = -\frac{\Lambda c^2}{8\pi G}}.$$

Hmm, sign check. The question writes $\rho' = \rho + \rho_V$ and $P' = P - \rho_V c^2$. Substitute $T'_{\mu\nu} = P' g_{\mu\nu} + (\rho' + P'/c^2)U_\mu U_\nu$:

$$T'_{\mu\nu} = (P - \rho_V c^2)g_{\mu\nu} + (\rho + P/c^2)U_\mu U_\nu = T_{\mu\nu} - \rho_V c^2 g_{\mu\nu}.$$

The original equation with primed source is:

$$R_{\mu\nu} - \frac{1}{2}g_{\mu\nu}R = -\frac{8\pi G}{c^4}T'_{\mu\nu} = -\frac{8\pi G}{c^4}T_{\mu\nu} + \frac{8\pi G}{c^2}\rho_V g_{\mu\nu}.$$

Compare with the modified equation rearranged:

$$R_{\mu\nu} - \frac{1}{2}g_{\mu\nu}R = -\frac{8\pi G}{c^4}T_{\mu\nu} - \Lambda g_{\mu\nu}.$$

Match coefficients of $g_{\mu\nu}$:

$$\frac{8\pi G}{c^2}\rho_V = -\Lambda \Rightarrow \boxed{\rho_V = -\frac{\Lambda c^2}{8\pi G}}.$$

(With the sign convention used in the question's Friedmann equations $\Lambda > 0$ produces accelerating expansion; a positive vacuum energy density corresponds to the vacuum equation of state $P_V = -\rho_V c^2$.)

Interpretation. ρ_V is the energy density of the *vacuum*: a Lorentz-invariant fluid with equation of state $P = -\rho c^2$, equivalent to a cosmological constant. Constant in space and time, indistinguishable from Λ in the field equations.

(d) **FRW identity** $\ddot{R}/R = H^2 + \dot{H}$

Problem. Prove $\ddot{R}/R = H^2 + \dot{H}$.

Definition $H = \dot{R}/R$ implies $\dot{R} = HR$. Differentiate w.r.t. t :

$$\ddot{R} = \dot{H}R + H\dot{R}.$$

Substitute $\dot{R} = HR$:

$$\ddot{R} = \dot{H}R + H^2R.$$

Divide by R :

$$\boxed{\frac{\ddot{R}}{R} = H^2 + \dot{H}.}$$

(e) **Differential equation for H**

Problem. For a flat Λ -matter universe show $2\dot{H} + 3H^2 = \Lambda c^2$.

For non-relativistic matter, $p = 0$. The two Friedmann equations reduce to

$$\frac{\ddot{R}}{R} = -\frac{4\pi G}{3}\rho + \frac{1}{3}\Lambda c^2,$$

$$H^2 = \frac{8\pi G}{3}\rho + \frac{1}{3}\Lambda c^2 \quad (k = 0).$$

Eliminate ρ . Multiply the second by $\frac{1}{2}$:

$$\frac{1}{2}H^2 = \frac{4\pi G}{3}\rho + \frac{1}{6}\Lambda c^2.$$

Add to the first:

$$\frac{\ddot{R}}{R} + \frac{1}{2}H^2 = \frac{1}{2}\Lambda c^2.$$

Use $()$:

$$\dot{H} + H^2 + \frac{1}{2}H^2 = \frac{1}{2}\Lambda c^2.$$

Multiply by 2:

$$\boxed{2\dot{H} + 3H^2 = \Lambda c^2.}$$

(f) **Solving for $H(t)$**

Problem. If $H \propto 1/t$ at early times, show $H = A \coth(3At/2)$ with $A^2 = \Lambda c^2/3$, and comment on late-time behaviour.

Early-time matter domination ignores Λ : $2\dot{H} + 3H^2 = 0$, $H = 2/(3t)$, indeed $\propto 1/t$. This sets the early-time boundary condition.

Rearrange $()$:

$$2 \frac{dH}{dt} = \Lambda c^2 - 3H^2 = 3(A^2 - H^2), \quad A^2 \equiv \Lambda c^2/3.$$

Separate variables:

$$\frac{2 dH}{A^2 - H^2} = 3 dt.$$

Use the hint $\int dx/(p^2 - x^2) = (1/p) \coth^{-1}(x/p)$ with $p = A$, $x = H$:

$$\frac{2}{A} \coth^{-1}(H/A) = 3t + \text{const.}$$

Apply early-time matching: as $t \rightarrow 0^+$, $H \rightarrow \infty$ so $\coth^{-1}(H/A) \rightarrow 0$, fixing the constant to 0:

$$\coth^{-1}(H/A) = \frac{3At}{2}.$$

Invert:

$$\boxed{H(t) = A \coth(3At/2), \quad A = \sqrt{\Lambda c^2/3}.}$$

Late time. $\coth(3At/2) \rightarrow 1$ as $t \rightarrow \infty$:

$$H \rightarrow A = \sqrt{\Lambda c^2/3}.$$

The Hubble parameter approaches a constant, the universe enters de Sitter (exponential) expansion $R \propto e^{At}$, dominated by the cosmological constant.

(g) Big rip from phantom vacuum $\rho_V \propto R^\epsilon$

Problem. If $\rho_V \propto R^\epsilon$ with $\epsilon > 0$, show that $R(t)$ diverges in finite time in a flat universe (matter neglected after t_0).

Friedmann equation, flat, matter neglected:

$$H^2 = \frac{8\pi G}{3} \rho_V = \frac{8\pi G}{3} \rho_{V,0} (R/R_0)^\epsilon.$$

Define the constant $B^2 \equiv (8\pi G/3) \rho_{V,0} R_0^{-\epsilon}$:

$$H^2 = B^2 R^\epsilon.$$

Take square root (expansion, $H > 0$):

$$\frac{\dot{R}}{R} = B R^{\epsilon/2}.$$

Rearrange:

$$R^{-1-\epsilon/2} dR = B dt.$$

Integrate from t_0 (where $R = R_0$):

$$\int_{R_0}^R R'^{-1-\epsilon/2} dR' = B(t - t_0).$$

Evaluate the left-hand integral with $\epsilon > 0$:

$$-\frac{2}{\epsilon} \left[R^{-\epsilon/2} - R_0^{-\epsilon/2} \right] = B(t - t_0).$$

Solve for $R^{-\epsilon/2}$:

$$R^{-\epsilon/2} = R_0^{-\epsilon/2} - \frac{1}{2}\epsilon B(t - t_0).$$

$R \rightarrow \infty$ corresponds to $R^{-\epsilon/2} \rightarrow 0$:

$$t_{\text{rip}} - t_0 = \frac{2R_0^{-\epsilon/2}}{\epsilon B} = \frac{2}{\epsilon} \frac{1}{H_0},$$

where $H_0 = BR_0^{\epsilon/2}$ is the Hubble rate at t_0 . This is finite for any $\epsilon > 0$:

$$\boxed{t_{\text{rip}} = t_0 + \frac{2}{\epsilon H_0} \quad (< \infty),}$$

i.e. the scale factor diverges at the finite future time t_{rip} — the “big rip” singularity of phantom dark energy.